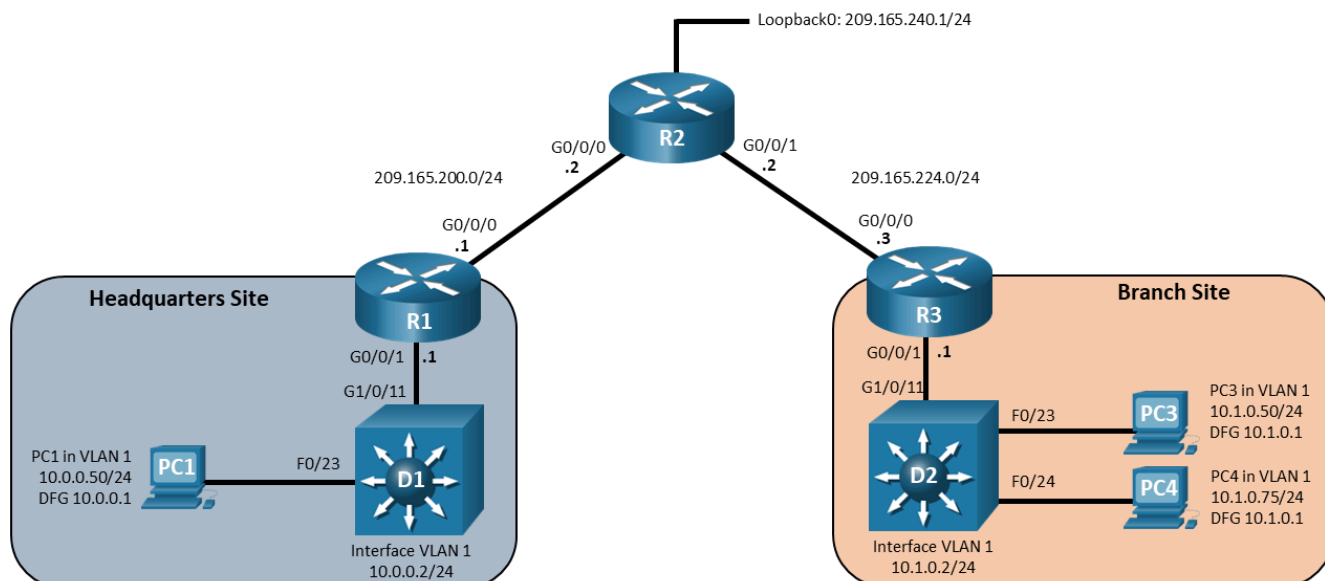


Lab - Implement NAT

Topology



Addressing Table

Device	Interface	IP Address	Default Gateway
R1	G0/0	209.165.200.1/24	N/A
	G0/1	10.0.0.1/24	
R2	G0/0	209.165.200.2/24	N/A
	G0/1	209.165.224.2/24	
	Loopback0	209.165.240.1/24	
R3	G0/0	209.165.224.3/24	N/A
	G0/1	10.1.0.1/24	
D1	VLAN 1	10.0.0.2/24	N/A
D2	VLAN 1	10.2.0.2/24	N/A
PC1	NIC	10.0.0.50/24	10.0.0.1
PC3	NIC	10.1.0.50/24	10.1.0.1
PC4	NIC	10.1.0.75/24	10.1.0.1

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Configure and Verify Static Inside NAT

Part 3: Configure and Verify Pooled NAT

Part 4: Configure and Verify NAT Overload

Background / Scenario

The HQ and Branch sites must be configured to support NAT. Specifically, the HQ and Branch routers will be configured to provide inside LAN users with outside public addresses using NAT. The HQ router will also provide static NAT to access the email server from the outside network.

Note: This lab is an exercise in configuring and verifying various methods of NAT and does not reflect networking best practices.

Note: The routers and switches used with CCNP hands-on labs are Cisco 4221 and Cisco 3650, both with Cisco IOS XE Release 16.9.4 (universalk9 image). Other routers and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the routers and switches have been erased and have no startup configurations. If you are unsure contact your instructor.

Required Resources

- 3 Routers (Cisco 4221 with Cisco IOS XE Release 16.9.4 universal image or comparable)
- 2 Switches (Cisco 3650 with Cisco IOS XE Release 16.9.4 universal image or comparable)
- 3 PCs (Choice of operating system with a terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet and serial cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each device.

- a. Console into each device, enter global configuration mode, and apply the basic settings. A command list for each device is provided below for the initial configurations.

Router R1

```
hostname R1
no ip domain lookup
line con 0
exec-timeout 0 0
logging synchronous
```

```
exit
line vty 0 4
  privilege level 15
  password cisco123
  exec-timeout 0 0
  logging synchronous
  login
exit
banner motd # This is R1, Implement NAT Lab #
interface g0/0
  ip address 209.165.200.1 255.255.255.0
  no shut
exit
interface g0/1
  ip address 10.0.0.1 255.255.255.0
  no shut
exit
ip route 0.0.0.0 0.0.0.0 g0/0 209.165.200.2
```

Router R2

```
hostname R2
no ip domain lookup
line con 0
  exec-timeout 0 0
  logging synchronous
exit
line vty 0 4
  privilege level 15
  password cisco123
  exec-timeout 0 0
  logging synchronous
  login
exit
banner motd # This is R2, Implement NAT Lab #
interface g0/0
  ip address 209.165.200.2 255.255.255.0
  no shut
exit
interface g0/1
  ip address 209.165.224.2 255.255.255.0
  no shut
exit
interface loopback 0
  ip address 209.165.240.1 255.255.255.0
  no shut
exit
```

```
ip route 10.0.0.0 255.255.255.0 g0/0 209.165.200.1  
ip route 10.1.0.0 255.255.255.0 g0/1 209.165.224.3
```

Router R3

```
hostname R3  
no ip domain lookup  
line con 0  
  exec-timeout 0 0  
  logging synchronous  
exit  
line vty 0 4  
  privilege level 15  
  password cisco123  
  exec-timeout 0 0  
  logging synchronous  
  login  
exit  
banner motd # This is R3, Implement NAT Lab #  
interface g0/0  
  ip address 209.165.224.3 255.255.255.0  
  no shut  
exit  
interface g0/1  
  ip address 10.1.0.1 255.255.255.0  
  no shut  
exit  
ip route 0.0.0.0 0.0.0.0 g0/0 209.165.224.2
```

Switch D1

```
hostname D1  
no ip domain lookup  
line con 0  
  exec-timeout 0 0  
  logging synchronous  
exit  
line vty 0 4  
  privilege level 15  
  password cisco123  
  exec-timeout 0 0  
  logging synchronous  
  login  
exit  
banner motd # This is D1, Implement NAT Lab #  
interface range g0/0-2  
  shutdown  
exit
```

```
interface range g0/0, g0/1
  no shutdown
exit
interface g0/1
  switchport mode access
  spanning-tree portfast
exit
interface vlan 1
  ip address 10.0.0.2 255.255.255.0
  no shutdown
ip default-gateway 10.0.0.1
```

Switch D2

```
hostname D2
no ip domain lookup
line con 0
  exec-timeout 0 0
  logging synchronous
exit
line vty 0 4
  privilege level 15
  password cisco123
  exec-timeout 0 0
  logging synchronous
  login
exit
banner motd # This is D2, Implement NAT Lab #
interface range g0/0-3
  shutdown
exit
interface range g0/0, g0/1-2
  no shutdown
exit
interface range g0/1-2
  switchport mode access
  spanning-tree portfast
exit
interface vlan 1
  ip address 10.1.0.2 255.255.255.0
  no shutdown
ip default-gateway 10.1.0.1
```

- b. Set the clock on each device to UTC time.
- c. Save the running configuration to startup-config.

Step 3: Verify reachability.

PC1 (10.0.0.50) should be able to ping PC3 (10.1.0.50), and PC4 (10.1.0.75) should be able to ping switch D1 (10.0.0.2). If not, notify your instructor so they may assist you in troubleshooting and correcting any incorrect configurations.

PC1 10.0.0.50

PC3 10.1.0.50

PC4 10.1.0.75

Part 2: Configure and Verify Static Inside NAT

In Part 2, you will configure and verify Static Inside NAT. The idea behind Static Inside NAT is to make an inside local address reachable via an outside global address. For this lab, we will make interface VLAN 1 on switch D1 appear as 209.165.200.99 on the outside network.

Step 1: On R1, configure Static Inside NAT.

Note: The following steps a and b do not have to be carried out in the order listed.

- a. Configure R1 to translate the address on D1 Interface VLAN 1 to 209.165.200.99.

```
R1(config)# ip nat inside source static 10.0.0.2 209.165.200.99
```

- b. On R1, specify the inside and outside interfaces for NAT purposes.

```
R1(config)# interface g0/0
R1(config-if)# ip nat outside
R1(config-if)# exit
R1(config)# interface g0/1
R1(config-if)# ip nat inside
R1(config-if)# exit
```

Step 2: Verify the NAT process is occurring on R1.

- a. On R1, issue the command **show ip nat translations**. In the output, you will see the static translation information.

```
R1# show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.200.99	10.0.0.2	---	---

Total number of translations: 1

- b. From the console of R2, send 10,000 pings to the destination address 209.165.200.99 using the command **ping 209.165.200.99 repeat 10000**. The pings should be successful.
- c. On R1, issue the command **show ip nat translations**. In the output, you will see the static translation as well as the translation used for the ping.

```
R1# show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.200.99	10.0.0.2	---	---
icmp	209.165.200.99:0	10.0.0.2:0	209.165.200.2:0	209.165.200.2:0

Total number of translations: 2

- d. From the console of R2, stop the ping if it is still running and then telnet to 209.165.200.99. You should be able to connect and login to D1. Use **cisco123** as the password when prompted.

- e. While logged in to D1, issue the command **show tcp brief**. In the output, you will see the addresses involved in the communication from D1's perspective.

```
D1# show tcp brief
```

TCB	Local Address	Foreign Address	(state)
054D9734	10.0.0.2.23	209.165.200.2.63955	ESTAB

- f. On R1, issue the command **show ip nat translations**. In the output, you will see the static translation as well as the translation used for the telnet session.

```
R1# show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.200.99	10.0.0.2	---	---
tcp	209.165.200.99:23	10.0.0.2:23	209.165.200.2:63955	209.165.200.2:63955

Total number of translations: 2

- g. Disconnect the telnet session in preparation for the next part of the lab.

Part 3: Configure and Verify Pooled NAT

In Part 3, you will configure, examine, and verify pooled NAT. Pooled NAT uses a pool of available outside addresses to dynamically translate inside addresses on a one-to-one basis. The drawback to Pooled NAT is that if you do not have a pool of addresses that is at least the same size as the number of addresses that need to be translated, there will be inside hosts that cannot send traffic outside until a translation times out or is manually cleared.

Step 1: On R3, configure Pooled NAT.

Note: The following steps a, b, c, and d do not have to be carried out in the order listed.

- a. On R3, create a standard access list that identifies the source addresses of traffic to be translated. Traffic with source addresses matching this access list are referred to as "interesting" and will be processed through the configured NAT rule. For this lab, specify the entire 10.1.0.0/24 network as interesting.

```
R3(config)# access-list 33 permit 10.1.0.0 0.0.0.255
```

- b. Next, create a pool of addresses to use for the interesting traffic translations. You should have authority to use these addresses, and outside networks should route traffic destined to these addresses to you. For this lab, we will use two addresses from the subnet connecting R3 and R2.

```
R3(config)# ip nat pool POOLEDNAT 209.165.224.5 209.165.224.6 prefix-length 24
```

- c. Associate the interesting addresses with the NAT pool.

```
R3(config)# ip nat inside source list 33 pool POOLEDNAT
```

- d. Establish the inside and outside interfaces on R3.

```
R3(config)# interface g0/0/0
R3(config-if)# ip nat outside
R3(config-if)# exit
R3(config)# interface g0/0/1
R3(config-if)# ip nat inside
R3(config-if)# exit
```

Step 2: Verify the NAT process is occurring on R3.

- a. On R3, issue the command **show ip nat translations**. The output will indicate that there are no translations.

- b. On R3, issue the command **show ip nat pool name POOLEDNAT**. The output will give you details about the pool that is available for translation.

```
R3# show ip nat pool name POOLEDNAT
```

NAT Pool Statistics

Pool name POOLEDNAT, id 1

	Assigned	Available
Addresses	0	2
UDP Low Ports	0	1024
TCP Low Ports	0	1024
UDP High Ports	0	129024
TCP High Ports	0	129024

(Low ports are less than 1024. High ports are greater than or equal to 1024.)

- c. From the command prompt on PC3, start a continuous ping to the destination address 209.165.224.2. The pings should be successful.
- d. On R3, issue the command **show ip nat translations**. In the output you will see the pooled translation as well as the translation used for the ping.

```
R3# show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.224.5	10.1.0.50	---	---
icmp	209.165.224.5:1	10.1.0.50:1	209.165.224.2:1	209.165.224.2:1

Total number of translations: 2

- e. From the command prompt on PC4, start a continuous ping to the destination address 209.165.224.2. The pings should be successful.

- f. On R3, issue the command **show ip nat translations**. In the output you will see the pooled translations as well as the translation used for the ping.

```
R3# show ip nat translations
```

Pro	Inside global	Inside local	Outside local	Outside global
---	209.165.224.6	10.1.0.75	---	---
---	209.165.224.5	10.1.0.50	---	---
icmp	209.165.224.6:6740	10.1.0.75:6740	209.165.224.2:6740	209.165.224.2:6740
icmp	209.165.224.5:1	10.1.0.50:1	209.165.224.2:1	209.165.224.2:1

Total number of translations: 4

- g. From the console of D2, send 10,000 pings to the destination address 209.165.224.2 using the command **ping 209.165.224.2 repeat 10000**. The pings should fail.
- h. On R3, you should see the following syslog message being repeated:

```
*Jan 25 16:52:01.498: %IOSXE-6-PLATFORM: R0/0: cpp_cp: QFP:0.0 Thread:000
TS:00000053243712630396 %NAT-6-ADDR_ALLOC_FAILURE: Address allocation failed; pool 1
may be exhausted [2]
```

We purposely created a pool of addresses that was too small to demonstrate one of the shortcomings of Pooled NAT. Verify the pool is exhausted by examining the output of the command **show ip nat pool name POOLEDNAT**.

```
R3# show ip nat pool name POOLEDNAT
```

NAT Pool Statistics

Pool name POOLEDNAT, id 1

	Assigned	Available
Addresses	2	0
UDP Low Ports	0	1024
TCP Low Ports	0	1024
UDP High Ports	0	129024
TCP High Ports	0	129024

(Low ports are less than 1024. High ports are greater than or equal to 1024.)

- Stop the pings on D2, PC3, and PC4. (**Note:** To stop the ping, press Ctrl+Shift+6.)
- In preparation for the next part of the lab, remove the pool and mapping commands. Leave the ACL and the interface specifications.

```
R3(config)# do clear ip nat translation *
R3(config)# no ip nat pool POOLEDNAT 209.165.224.5 209.165.224.6 prefix-length 24
R3(config)# no ip nat inside source list 33 pool POOLEDNAT
```

Part 4: Configure and Verify NAT Overload

In Part 4, you will configure and examine NAT Overload, which is also called Port Address Translation or PAT. PAT overloads on an interface address or on a pool of addresses by associating each traffic session with a unique set of port numbers. The advantage, especially for smaller networks, is that everyone can get out of the network by leveraging a single outside IP address. The drawback to PAT is that you have a limit of about 65,000 to the number of translations possible via a single address. The impact of this limit is felt in large networks, where you must have multiple outside IP addresses to overload onto to ensure everyone can communicate.

As a reminder, R3 still has interfaces setup as outside and inside, and still has access list 33 specifying the LAN network address. You will leverage those existing configurations to complete this part of the lab.

For this lab, R3 will overload on the interface specified as outside, which is g0/0/0.

- Configure the NAT statement directing the router to translate addresses matching access list 33 to the outside interface IP address, keying on port numbers.

```
R3(config)# ip nat inside source list 33 interface g0/0/0 overload
```

- On R3, issue the command **show ip nat translations**. The output will indicate that there are no translations.
- From the command prompt on PC3, start a continuous ping to the destination address 209.165.224.2. The pings should be successful.
- On R3, issue the command **show ip nat translations**. In the output you will see the PAT translation used for the ping.

```
R3# show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 209.165.224.3:1     10.1.0.50:1       209.165.224.2:1    209.165.224.2:1
Total number of translations: 1
```

- From the command prompt on PC4, start a continuous ping to the destination address 209.165.224.2. From the console of D2, send 100,000 pings to the same address. The pings should be successful.
- On R3, issue the command **show ip nat translations**. In the output you will see the PAT translations as well as the translation used for the ping from D2 and PC4.

R3# **show ip nat translations**

```
Pro  Inside global      Inside local      Outside local      Outside global
icmp 209.165.224.3:6791 10.1.0.75:6791   209.165.224.2:6791 209.165.224.2:6791
icmp 209.165.224.3:1   10.1.0.50:1      209.165.224.2:1    209.165.224.2:1
icmp 209.165.224.3:6784 10.1.0.2:1       209.165.224.2:1    209.165.224.2:6784
Total number of translations: 3
```

- e. Stop the continuous pings on D2, PC3, and PC4.

Router Interface Summary Table

Router Model	Ethernet Interface #1	Ethernet Interface #2	Serial Interface #1	Serial Interface #2
1800	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2801	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
2811	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
4221	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
4300	Gigabit Ethernet 0/0/0 (G0/0/0)	Gigabit Ethernet 0/0/1 (G0/0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.